EXPLORATORY DATA ANALYSIS REPORT

E-Commerce Order Dataset — Project 2

Industrial Training Kit | Data Analytics | DecodeLabs

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Dataset: Dataset_for_Data_Analytics.xlsx (1,200 records × 14 fields)

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1. Problem Statement & Objective

DecodeLabs provided a transactional e-commerce order dataset containing 1,200 orders placed between January 1, 2023 and June 30, 2025. As a Data Analyst, the objective of this project is to perform a complete Exploratory Data Analysis (EDA) on this dataset to:

- Calculate basic descriptive statistics (mean, median, count, spread) for all key numeric fields.
- Identify trends across time, product categories, payment methods and customer acquisition channels.
- Detect and interpret outliers — distinguishing genuine data-entry noise from legitimate high-value “signal” orders.
- Summarize key business observations and translate statistical findings into actionable insight (the “So What?” test).

This report follows the IPO framework (Input → Process → Output): the raw order data is the evidence, the statistical and visual analysis is the investigation, and the findings below are the verdict.

2. Methodology

The analysis was carried out in Python using pandas, NumPy and Matplotlib. The workflow followed five steps:

- Data Loading & Structure Check — loaded the single-sheet workbook and verified column data types.
- Data Quality / Forensics — checked for missing values, duplicate orders, invalid dates, and arithmetic consistency (TotalPrice = Quantity × UnitPrice).
- Univariate Analysis — computed the five-number summary (min, Q1, median, Q3, max), mean, and standard deviation for every numeric column; assessed distribution shape using skewness.
- Outlier Detection — applied both the IQR method (flagging values beyond Q1 − 1.5×IQR or Q3 + 1.5×IQR) and the Z-score method (|Z| > 3) to cross-validate outliers.
- Trend & Relationship Analysis — grouped data by year, product, payment method, order status and referral source; computed the Pearson correlation matrix between numeric fields.

3. Dataset Overview & Data Quality Assessment

The dataset contains 1,200 order-level records across 14 fields, covering order identifiers, dates, customer IDs, product details, pricing, payment methods, fulfillment status, and marketing attribution (coupon code and referral source).

3.1 Structure

Field	Type	Description
OrderID	Text	Unique order identifier (1,200 unique values, no duplicates)
Date	Date	Order date (Jan 1, 2023 – Jun 30, 2025)
CustomerID	Text	Customer identifier (1,189 unique customers)
Product	Category	One of 7 product types

Field	Type	Description
Quantity	Integer	Units ordered (1–5)
UnitPrice	Decimal	Price per unit ($11.39 – $699.93)
ShippingAddress	Text	Placeholder address field
PaymentMethod	Category	One of 5 payment types
OrderStatus	Category	One of 5 fulfillment states
TrackingNumber	Text	Shipment tracking ID
ItemsInCart	Integer	Items in cart at checkout (1–10)
CouponCode	Category	Coupon applied (or blank if none)
ReferralSource	Category	Marketing channel (5 channels)
TotalPrice	Decimal	Order value = Quantity × UnitPrice

3.2 Data Quality Findings

Following the "dataset as digital evidence" framework, the data was inspected for the four classic integrity issues: missing values, invalid formats, misalignment, and corruption. The dataset proved to be clean and analysis-ready:

- Completeness: 13 of 14 fields are 100% populated. The only field with nulls is CouponCode, with 309 blank entries (25.75%) — these represent orders where no coupon was used, not data corruption.
- No duplicate records: 0 duplicate OrderIDs and 0 fully duplicated rows out of 1,200.
- Arithmetic integrity verified: TotalPrice exactly equals Quantity × UnitPrice for all 1,200 rows (0 mismatches), confirming the pricing logic is internally consistent.
- Valid ranges: no negative or zero values in Quantity, UnitPrice, ItemsInCart, or TotalPrice; all dates parse correctly with no invalid-format entries.
- Low-value field: ShippingAddress contains a placeholder pattern ("### Main St", street-number range 100–999) and carries no real geographic signal, so it was excluded from trend analysis.

Data Completeness Snapshot

Total Valid Records: 1,200 / 1,200 (100%)

Fields with Missing Values: 1 of 14 (CouponCode — explainable, not an error)

Format Errors: 0% | Duplicate Records: 0% | Calculation Errors: 0%

4. Descriptive Statistics (Five-Number Summary)

Basic statistics were calculated for all four numeric fields. Quantity and UnitPrice (and therefore ItemsInCart) are close to symmetrical, while TotalPrice — a product of two variables — is right-skewed, which is the expected mathematical result of multiplying two roughly uniform distributions.

Metric	Quantity	Unit Price ($)	Items In Cart	Total Price ($)
Count	1,200	1,200	1,200	1,200
Mean	2.95	356.41	5.49	1,053.97
Median	3.00	364.21	5.00	823.62
Std. Deviation	1.41	197.18	2.28	819.86
Minimum	1	11.39	1	11.39
Q1 (25th pct.)	2	186.06	4	410.52
Q3 (75th pct.)	4	521.57	7	1,578.48
Maximum	5	699.93	10	3,456.40
Skewness	+0.03 (symmetric)	−0.03 (symmetric)	+0.00 (symmetric)	+0.89 (right-skewed)

4.1 Distribution Shape

Because TotalPrice is right-skewed, the median ($823.62) is a more representative “typical order value” than the mean ($1,053.97) — the mean is pulled upward by a tail of large-basket, high-unit-price orders. This mirrors the textbook “Mean vs. Median” principle: use the median for skewed monetary data such as order values, exactly as one would for salaries or house prices.

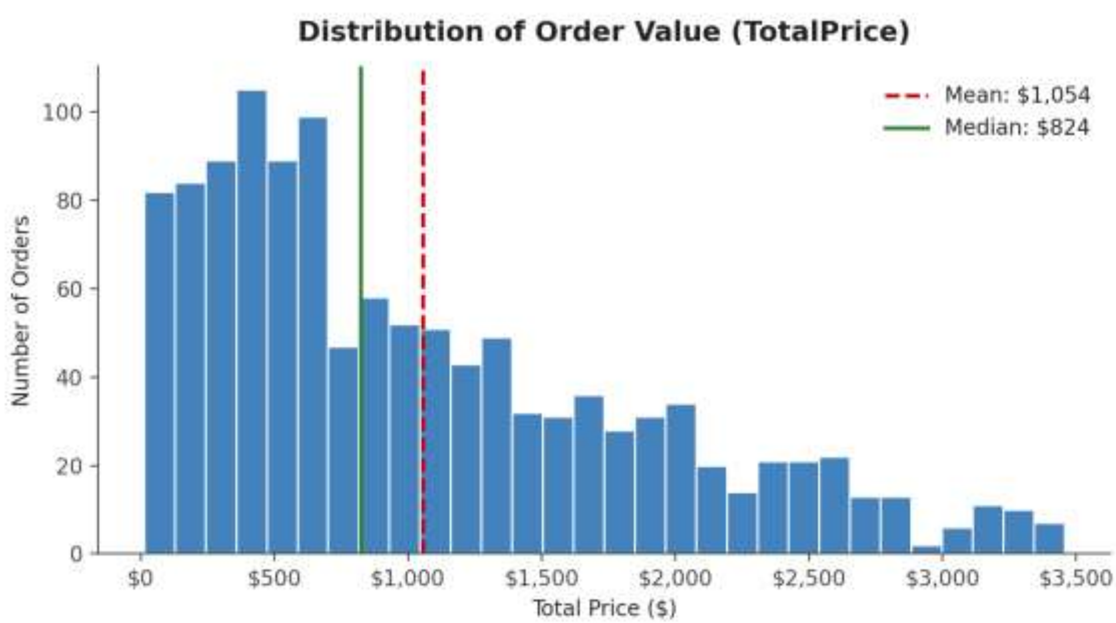

Figure 1. Distribution of Order Value (TotalPrice). The right tail confirms positive skew (0.89); the median sits left of the mean.

5. Outlier Detection

Two complementary outlier-detection methods were applied to TotalPrice, the dataset's key monetary field, to cross-check results:

- IQR Method: Q1 = $410.52, Q3 = $1,578.48, IQR = $1,167.96. Outlier bounds = [−$1,341.41, $3,330.41]. Since TotalPrice cannot be negative, only the upper bound is meaningful → 8 orders (0.67% of all orders) exceed $3,330.41.
- Z-Score Method: Using |Z| > 3 as the threshold, 0 orders are flagged as statistical outliers on TotalPrice.

The two methods disagree because the Z-score method assumes a roughly normal distribution and is far less sensitive on skewed data, whereas the IQR method is robust to skew — exactly the trade-off described in the IQR-vs-Z-Score framework. For this 'dirty', real-world-style business data, the IQR method is the more trustworthy of the two.

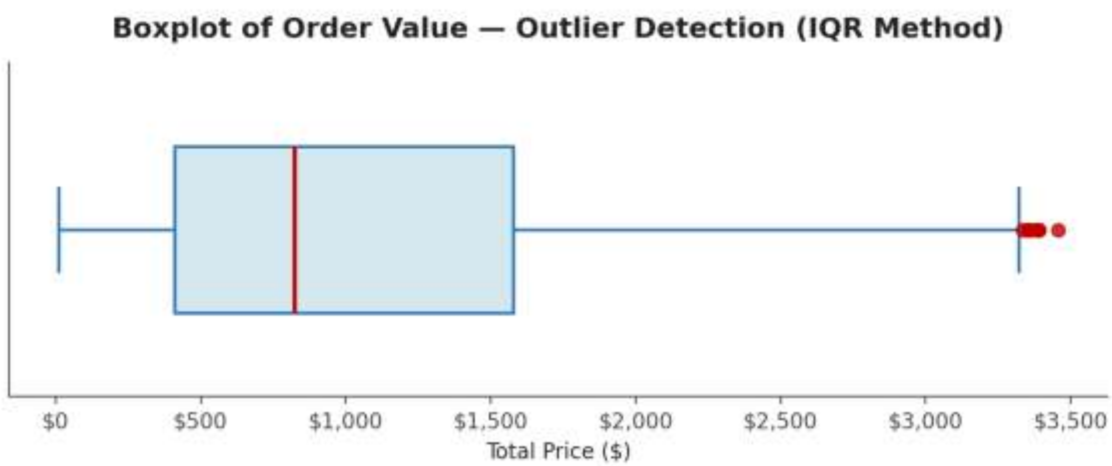

Figure 2. Boxplot of Order Value. Red dots mark the 8 orders flagged by the IQR method.

5.1 Are These Outliers Noise or Signal?

Inspecting the 8 flagged orders shows a consistent pattern: every one of them has the maximum possible Quantity (5 units) combined with a high UnitPrice (~$667–$691). There are no impossible values, broken formatting, or logic errors — TotalPrice still equals Quantity × UnitPrice exactly in every case.

OrderID	Product	Quantity	Unit Price ($)	Total Price ($)	Order Status
ORD200107	Printer	5	670.75	3,353.75	Shipped
ORD200326	Laptop	5	670.48	3,352.40	Returned
ORD200328	Tablet	5	674.04	3,370.20	Cancelled
ORD200469	Chair	5	676.98	3,384.90	Cancelled
ORD200632	Laptop	5	678.16	3,390.80	Delivered
ORD200789	Tablet	5	691.28	3,456.40	Delivered
ORD201065	Printer	5	666.80	3,334.00	Delivered
ORD201122	Monitor	5	678.19	3,390.95	Returned

Verdict: these are SIGNAL, not noise. They represent the dataset's genuine large-basket, high-value orders (bulk purchases of premium electronics/furniture) rather than data-entry mistakes. They should be investigated as a VIP / bulk-order segment, not deleted or capped.

6. Trend Analysis

6.1 Revenue by Product Category

Revenue is fairly evenly spread across the seven product categories ($151.7K–$195.6K each), with no single category dominating. Chair and Printer are the top two revenue generators despite Chair having a lower average unit price, driven by stronger order volume.

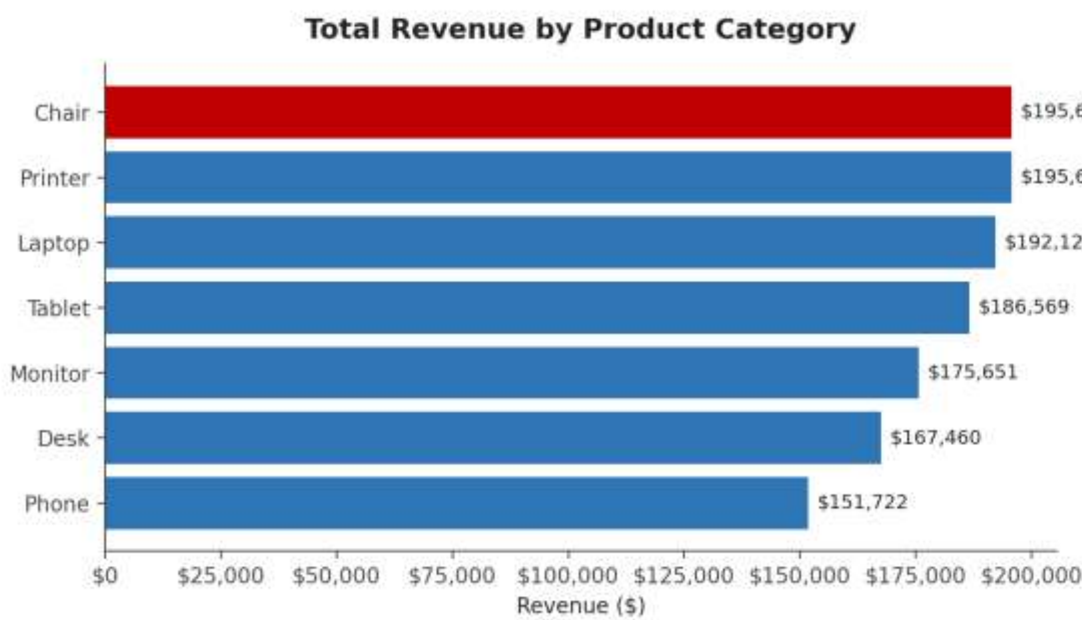

Figure 3. Total Revenue by Product Category.

Product	Orders	Revenue ($)	Avg. Unit Price ($)	Avg. Quantity
Chair	178	195,620.11	355.66	3.16
Printer	181	195,612.61	351.71	2.99
Laptop	173	192,126.56	357.71	3.09
Tablet	179	186,568.95	367.68	2.78
Monitor	163	175,651.41	358.66	2.94
Desk	170	167,459.93	329.61	2.99
Phone	156	151,722.39	375.22	2.63

6.2 Order Fulfillment Status — The Most Important Finding

Orders are nearly evenly split across five statuses, but combining Cancelled (250 orders, 20.8%) and Returned (247 orders, 20.6%) shows that 497 orders — 41.4% of all orders — never convert into a kept sale.

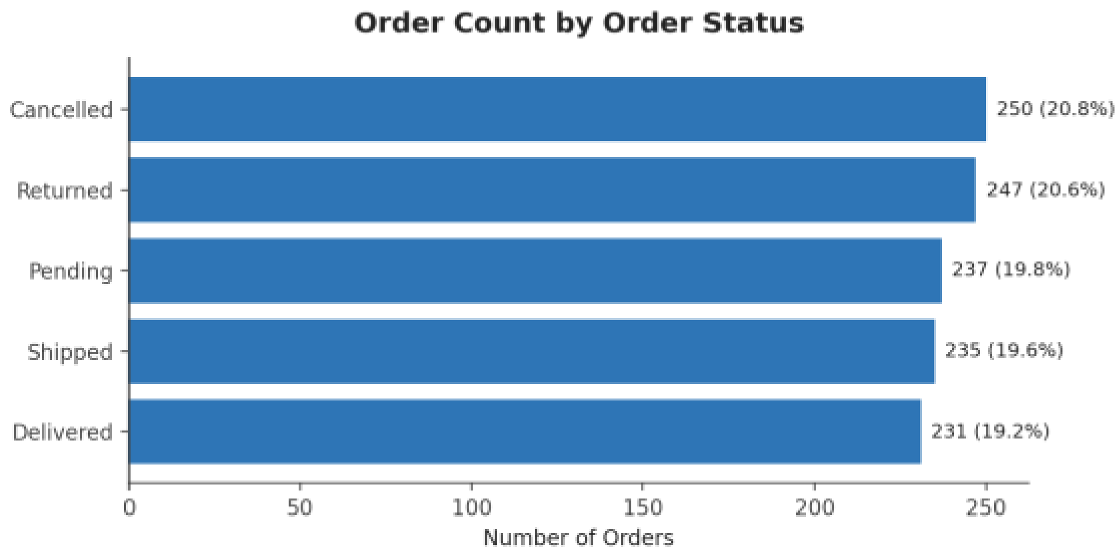

Figure 4. Order Count by Order Status.

> **Business Impact — Translating the Data into a Dollar Figure**
>
> Combined Cancelled + Returned orders: 497 (41.4% of all 1,200 orders)
>
> Revenue tied up in Cancelled/Returned orders: $519,673.91 — about 41.1% of total recorded revenue ($1,264,761.96)
>
> So What: Nearly 2 out of every 5 dollars of 'recorded' revenue never actually lands as completed sales. This is the single highest-leverage finding in the dataset and should be the headline of any executive summary built from this data.

6.3 Time Trend

Order volume by calendar month appears higher in Jan–Jun than in Jul–Dec when looking at the raw totals — but this is a data-coverage artifact, not seasonality, the classic ‘confounder’ trap. The dataset includes full years for 2023 and 2024 but only six months (Jan–Jun) of 2025. Once the years are separated, no consistent seasonal pattern emerges in either 2023 or 2024.

Figure 5. Monthly Order Volume Split by Year. 2025 (dashed) stops in June, confirming the partial-year effect rather than a true seasonal trend.

Year	Orders	Revenue ($)	Avg. Order Value ($)	Coverage
2023	510	552,643.24	1,083.61	Full year (Jan–Dec)
2024	459	480,235.87	1,046.27	Full year (Jan–Dec)
2025	231	231,882.85	1,003.82	Partial year (Jan–Jun only)

Annualizing 2025 (231 orders over 6 months → ~462 projected for the full year) shows order volume is essentially flat year-over-year, not declining — a materially different conclusion than a naive look at the raw totals would suggest.

6.4 Payment Method & Referral Source

Both fields are also fairly evenly distributed, suggesting the business is not over-reliant on a single channel:

- Payment Method: Credit Card leads with $263,847.63 (234 orders), followed closely by Online ($262,442.94) and Cash ($259,786.29); Debit Card is lowest at $232,361.18.
- Referral Source: Instagram drives the most revenue ($275,285.45, 259 orders), followed by Email ($261,808.55); Referral is the smallest channel ($226,815.58).

- Coupons: 891 of 1,200 orders (74.25%) used a coupon. Orders with a coupon actually have a slightly higher average value ($1,057.64) than those without ($1,043.37), so coupons show no evidence of cannibalizing order size in this dataset.

7. Correlation Analysis

The Pearson correlation coefficient (r) was calculated between all numeric fields to map the strength and direction of their linear relationships.

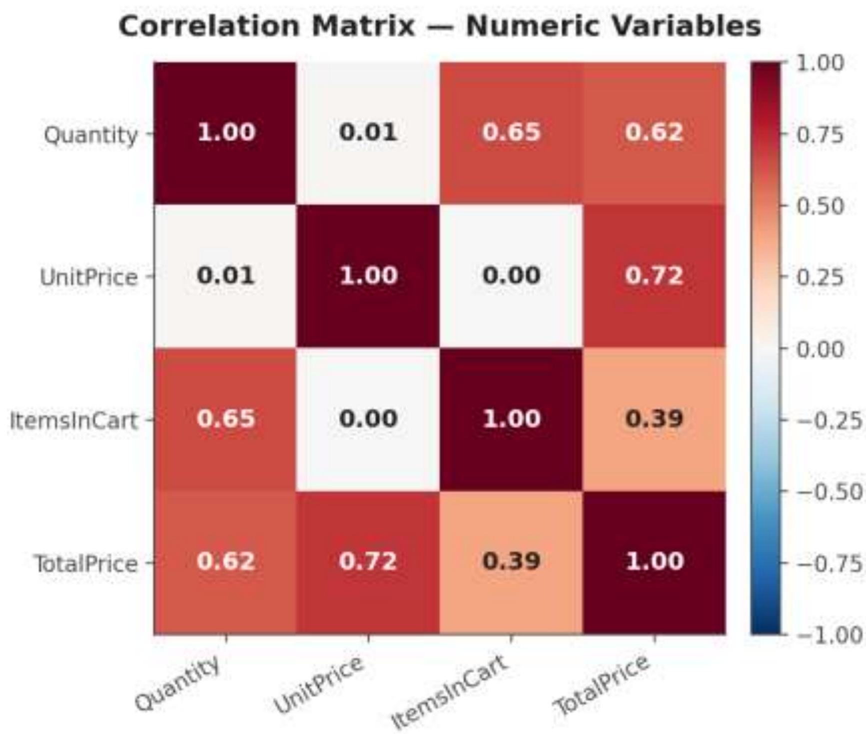

Figure 6. Correlation Matrix — Numeric Variables.

- UnitPrice ↔ TotalPrice: r = 0.72 (strong positive) — the price of the item is the single biggest driver of order value.
- Quantity ↔ TotalPrice: r = 0.62 (moderate-strong positive) — buying more units also pushes order value up, but less strongly than unit price does.
- Quantity ↔ ItemsInCart: r = 0.65 (moderate-strong positive) — customers who add more items to their cart tend to also order more units, as expected.
- Quantity ↔ UnitPrice: r = 0.01 (no correlation) — customers do not systematically buy more units of cheaper items or vice-versa.

The Golden Rule: Correlation ≠ Causation

These correlations are clues, not verdicts. For example, UnitPrice and TotalPrice are correlated by construction (TotalPrice = Quantity × UnitPrice), not because of any external business behavior — always check whether a relationship is mathematically built-in before treating it as a discovered insight.

8. Key Observations & Business Insights

What the Data Says	Business Diagnosis
41.4% of orders end in Cancelled or Returned status, tying up $519,673.91 in revenue.	This is the most urgent issue in the dataset. Even a 10-percentage-point reduction in cancellations/returns would protect roughly $125,000 in revenue — investigate root causes (payment failures, fulfillment delays, product fit) before any growth initiative.
TotalPrice is right-skewed (skew = 0.89); mean ($1,053.97) > median ($823.62).	Report the median, not the mean, when describing a 'typical' order — the mean overstates what most customers actually spend.
8 orders (0.67%) are statistical outliers, all driven by max-quantity (5 units) + high-unit-price combinations, not data errors.	These are a legitimate bulk/high-value order segment worth a dedicated VIP or wholesale offer, not a data-cleaning target.
UnitPrice (r=0.72) influences order value more than Quantity (r=0.62).	Pricing strategy and premium product mix have more leverage on revenue than encouraging larger cart sizes.
Revenue is evenly spread across 7 products, 5 payment methods and 5 referral sources (no channel >23% share).	The business is well-diversified with no single point of failure — marketing spend and inventory can stay broadly balanced rather than over-indexing on one channel.
Apparent Jan–Jun order-volume spike is a partial-2025-data artifact, not seasonality.	Avoid building a seasonal marketing calendar from this raw total — re-run the trend once full-year 2025 data is available.

9. Conclusion & Recommendations

This EDA confirms the dataset is clean, internally consistent, and free of formatting or duplication errors, allowing the analysis to focus entirely on business signal rather than data forensics. The headline finding — a 41.4% combined cancellation/return rate representing roughly $519.7K in unrealized revenue — is the clearest ‘so what’ in the dataset and should anchor any follow-up investigation or dashboard.

Recommendations

- Prioritize a root-cause analysis of Cancelled and Returned orders (e.g., by payment method or product) before investing in customer acquisition.
- Use the median, not the mean, when communicating typical order value to stakeholders, given the confirmed right skew.
- Treat the 8 high-value orders as a candidate VIP/bulk-buyer segment for targeted retention, rather than as data to be cleaned or removed.
- Re-evaluate the monthly trend once full 2025 data is available, since the current Jan–Jun skew is a data-coverage artifact.
- Maintain the current channel diversification across products, payment methods and referral sources, as no single channel is a structural weak point today.

“EDA is the diagnostic phase for data integrity — the analyst's job is to translate noise into signal, and signal into a decision.”